Supplementary Material to "A Multidimensional Agent-Based Modeling and Simulation Framework for Urban Water Demand Forecast under Socioeconomic Inequality"

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Abstract

Traditional aggregate models fail to capture complex interactions between spatial heterogeneity and human behavior in urban water demand forecasting. We present a multidimensional Agent-Based Model for Water Prediction (ABM-WP) framework to simulate neighborhood-scale water consumption dynamics under socioeconomic inequality data from Salvador, Brazil. A key contribution is the systematic analysis of multiple 18 configuration scenarios integrating the Environmentalist, Moderate, and Wasteful behavioral profiles with socioeconomic income growth and population expansion. ABM-WP on the GAMA platform was compared to statistical seasonal ARIMA and machine learning models (LSTM, MLP, SVR). ANOVA and Tukey tests revealed significant tipping points where socioeconomic catalysts cause demand to deviate from historical trends over 220,000 m^3 annually. ABM-WP identifies critical peaks that data-based methods fail to predict. Spatial consumption patterns analysis is driven by connection density and socioeconomic clustering. ABM-WP is a robust decision-support tool for governance, providing insights for urban sustainability targeting conservation and infrastructure planning.

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[1] Research conducted while a visiting professor at the Federal University of Bahia (UFBA).

1. Performance Measures, Environment, Actuators, and Sen- sors (PEAS) to describe the task environment of the AA, PA, and CA agents.

Table 1: PEAS description for the AA, responsible for data collection, preprocessing, and analysis. Source: Elaborated by the author.

PEAS	Description
Performance	Quantity and quality of data processed per consumer unit
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Analyze family-grouped data Compare and process consumption Send processed data to predictor agent
Sensors	Residence geolocation Residence types and profile Population growth

Table 2: PEAS description for the PA, responsible for making scenario simulations. Source: Elaborated by the author.

PEAS	Description
Performance	Maintain prediction quality with minimal Mean Absolute Percentage Error (MAPE)
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Select optimal prediction model Generate family consumption forecasts Send forecasts to AC
Sensors	Data provided by AA

Table 3: PEAS description for the CA, responsible for consolidating the generated forecasts. Source: Elaborated by the author.

PEAS	Description
Performance	Number of reports with consumption forecasts issued
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Send water demand reports Distribute to PA water demand reports
Sensors	Data generated by the PA

2. Architectural Design

The architectural model of the ABM-WP framework was structured using the Tropos methodology, which details the interaction between human actors and the agents responsible for forecasting the water demand. The Operator is the central point of progression, managing data dependencies with the Family (household data provider) and the Meter (responsible for recording monthly and historical consumption). Operational intelligence is distributed across three specialized agents: the Analyzer Agent, responsible for analyzing, preprocessing, and consolidating customer data; the Predictor Agent, generating individualized consumption forecasts per family and per scenario; and the Communicator Agent, which prepares and consolidates scenario predictions. The workflow is driven by dependencies and objectives.

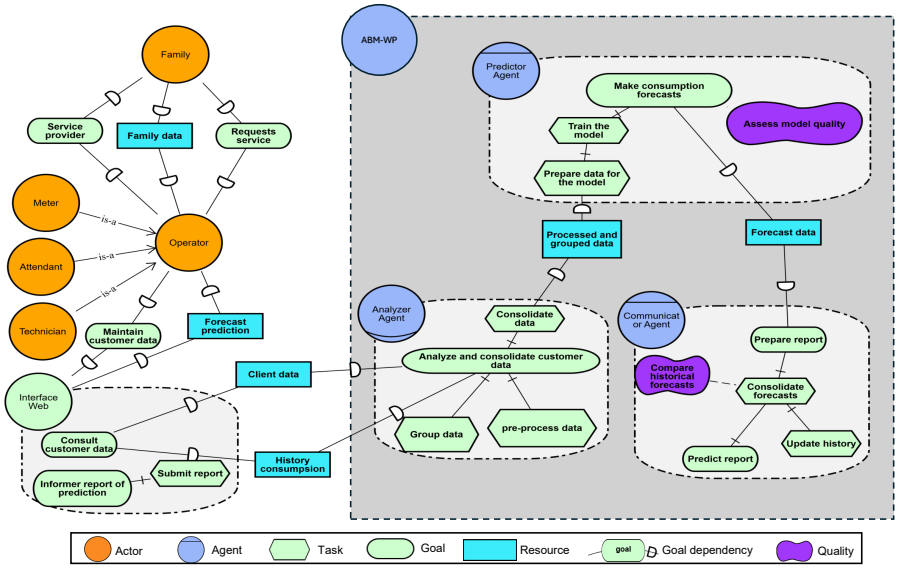

Figure 1: Architecture of the ABM-WP framework, encompassing the interaction between actors, tasks and agents.

3. Consumption Profiles by Sector

Ref.: January/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S027	658	314	76	268	145.80	47.72	11.55	40.73
S029	582	319	49	214	154.32	54.81	8.42	36.77
S023	554	316	38	200	131.63	57.04	6.86	36.10
S048	489	296	54	139	104.16	60.53	11.04	28.43
S003	453	292	33	128	104.80	64.46	7.28	28.26
S024	445	248	30	167	135.99	55.73	6.74	37.53
S020	413	206	30	177	141.14	49.88	7.26	42.86
S078	403	220	43	140	125.34	54.59	10.67	34.74
S079	394	223	35	136	107.66	56.60	8.88	34.52
S032	354	220	40	94	104.59	62.15	11.30	26.55
S099	351	152	37	162	147.69	43.30	10.54	46.15
S080	348	206	49	93	91.65	59.20	14.08	26.72
S067	343	211	28	104	111.11	61.52	8.16	30.32
S021	334	224	34	76	85.51	67.07	10.18	22.75
S000	323	232	25	66	77.44	71.83	7.74	20.43
S008	302	72	23	207	188.08	23.84	7.62	68.54
S026	300	195	25	80	96.62	65.00	8.33	26.67
S086	295	158	37	100	130.44	53.56	12.54	33.90
S063	274	153	22	99	117.77	55.84	8.03	36.13
S033	273	151	26	96	113.87	55.31	9.52	35.16
S007	272	78	18	176	160.73	28.68	6.62	64.71
S034	268	125	27	116	168.65	46.64	10.07	43.28
S005	267	195	22	50	85.48	73.03	8.24	18.73
S025	254	152	21	81	119.07	59.84	8.27	31.89
S073	238	154	21	63	87.61	64.71	8.82	26.47
S081	234	143	17	74	104.89	61.11	7.26	31.62
S093	228	178	9	41	61.78	78.07	3.95	17.98
S074	222	152	19	51	97.31	68.47	8.56	22.97
S077	215	97	22	96	223.85	45.12	10.23	44.65
S092	210	125	7	78	121.78	59.52	3.33	37.14
S028	209	120	19	70	244.10	57.42	9.09	33.49
S018	190	140	7	43	66.58	73.68	3.68	22.63
S072	188	65	17	106	315.92	34.57	9.04	56.38
S084	186	108	13	65	115.18	58.06	6.99	34.95

Ref.: January/2026

CD_SETOR	total_consumers	environmentalist_count	moderate_count	wasteful_count	avg_daily_consumption	pct_environmentalist	pct_moderate	pct_wasteful
S022	183	80	17	86	150.74	43.72	9.29	46.99
S060	173	64	20	89	230.03	36.99	11.56	51.45
S066	173	108	15	50	104.44	62.43	8.67	28.90
S082	173	125	14	34	76.27	72.25	8.09	19.65
S019	172	73	19	80	175.34	42.44	11.05	46.51
S083	163	116	13	34	83.41	71.17	7.98	20.86
S004	162	104	12	46	93.19	64.20	7.41	28.40
S036	160	76	16	68	182.12	47.50	10.00	42.50
S047	158	104	15	39	87.81	65.82	9.49	24.68
S089	157	112	12	33	87.37	71.34	7.64	21.02
S015	153	99	15	39	93.24	64.71	9.80	25.49
S017	151	111	10	30	65.02	73.51	6.62	19.87
S001	146	67	21	58	124.41	45.89	14.38	39.73
S044	138	43	14	81	353.31	31.16	10.14	58.70
S006	138	92	16	30	85.21	66.67	11.59	21.74
S091	136	59	15	62	171.26	43.38	11.03	45.59
S031	127	43	8	76	389.42	33.86	6.30	59.84
S075	118	46	14	58	155.11	38.98	11.86	49.15
S035	105	59	8	38	132.70	56.19	7.62	36.19
S030	103	27	6	70	278.62	26.21	5.83	67.96
S045	97	21	8	68	347.00	21.65	8.25	70.10
S068	91	21	2	68	445.09	23.08	2.20	74.73
S095	87	60	13	14	77.27	68.97	14.94	16.09
S039	83	32	10	41	144.35	38.55	12.05	49.40
S088	83	39	8	36	157.17	46.99	9.64	43.37
S046	79	35	6	38	210.04	44.30	7.59	48.10
S094	77	51	6	20	89.96	66.23	7.79	25.97
S040	66	24	5	37	210.93	36.36	7.58	56.06
S013	23	11	2	10	134.02	47.83	8.70	43.48
S002	1	0	0	1	71778.69	0.00	0.00	100.00

4. Shapiro-Wilk Normality Test

Table 4: Shapiro-Wilk normality test results for water demand scenarios showing the test statistic (W), *p*-values, and normality assessment for each consumption scenario. Source: Elaborated by the author.

Scenario	**W statistic**	**p-value**	**Normal Distribution**
CI	0.9356	0.5052	Yes
CII	0.9356	0.5052	Yes
CIII	0.9356	0.5052	Yes
CIV	0.9583	0.7666	Yes
CV	0.9585	0.7687	Yes
CVI	0.9587	0.7709	Yes
CVII	0.9583	0.7668	Yes
CVIII	0.9585	0.7689	Yes
CIX	0.9587	0.7711	Yes
CX	0.9582	0.7649	Yes
CXI	0.9583	0.7667	Yes
CXII	0.9585	0.7684	Yes
CXIII	0.9584	0.7671	Yes
CXIV	0.9586	0.7693	Yes
CXV	0.9588	0.7717	Yes
CXVI	0.8935	0.1857	Yes
CXVII	0.9136	0.3069	Yes
CXVIII	0.9370	0.5205	Yes
LSTM	0.6207	0.0001	No

5. Expert Validation Questionnaire

EXPERT VALIDATION QUESTIONNAIRE (EXPERT JUDGMENT)

Project: Agent-Based Modeling for Water Demand Forecasting in Salvador

Dear Specialist,

This form aims to conduct **Face Validity** and **Operational Validity** of the computational model developed to simulate water demand in the Itapuã neighborhood. Your practical experience in the operation and management of the water supply system is essential to confirm whether the behavior of the model and its results reflect the reality of the sector.

The responses will be used to calculate the Content Validity Index (CVI) of the study. Please use the provided numerical scale to indicate your degree of agreement or the representativeness of each item.

To assist in understanding the research and answering the questionnaire, we have attached (click here to download):

1 - Hypsometric map showing the distribution of consumption profiles;

2 - Report on the distribution of consumption profiles by sector;

3 - Graph of the simulation results for the 18 scenarios;

4 - Report with the consolidated simulation results by year and scenario.

SECTION 1: SPECIALIST PROFILE

1. Full Name: _____
2. Current Position/Role: _____
3. Years of experience in sanitation/water supply: [ ] years.
4. Years of experience/specific knowledge about the Itapuã region: [ ] years.

SECTION 2: CONCEPTUAL AND BEHAVIORAL VALIDATION

5. The study classifies consumers into three profiles: **Environmentalist** (<100 L/cap/day), **Moderate** (100-121.5 L/cap/day), and **Waster** (>121.5 L/cap/day). **Can the proposed segmentation be used to represent the reality of local consumers?**

[1] Not Representative [2] Slightly Representative [3] Neutral / I can't say

[4] Representative [5] Very Representative

6. Socioeconomic Influence: The model assumes that income and population growth factors (e.g., the head of household increases their monthly income, or the number of

residents grows) influence changes in consumption behavior over time. **How relevant is this factor to the dynamics of the neighborhood?**

[1] Irrelevant [2] Slightly Relevant [3] Neutral / I can't say

[4] Relevant [5] Very Relevant

SECTION 3: OPERATIONAL VALIDATION (RESULTS)

7. Spatial Distribution (Hotspots): The simulation identified critical concentrations of consumption in census sectors S027, S029, S023, S048, and S003. **Was the model accurate in identifying these areas as critical (high demand or losses)?**

[1] Inaccurate [2] Slightly Accurate [3] Fair [4] Accurate [5] Very Accurate

8. Relationship with Topography: The model concluded that there is no significant correlation between altitude and consumed volume in Itapuã (i.e., topography does not define consumption in this area). **Do you agree with the validity of this conclusion for the region?**

[1] Disagree [2] Partially Disagree [3] Neither Agree / Neither Disagree

[4] Partially Agree [5] Agree

9. Applicability: Considering the breadth of the 18 simulated scenarios, to what extent do you consider this set of projections to constitute a robust basis for contingency planning and decision support in water demand management in Itapuã?

[1] Not Applicable (Not useful for supporting decisions) [2] Low Applicability

[3] Neutral / I can't say [4] Useful for Planning

[5] High Applicability (Essential for decision support)

SECTION 4: UTILITY AND APPLICABILITY

10. In your technical opinion, is this Agent-Based Model (ABM) a useful tool to support infrastructure planning and demand management in Salvador?

[1] Useless [2] Slightly Useful [3] Neutral / Indifferent

[4] Useful [5] Very Useful

11. Space for suggestions or final observations:

_____

_____

Signature:: _____ **Date:** __/__/____